Lecture #7: The Derivative Function & Interpretations

MATH 145 – Calculus I

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Example: The Derivative at a Point

Example: Find $f'(2)$ for $f(x) = x^2$.

The Derivative Function

What if we do this for all x at once, symbolically, instead of just $x = 2$?

This leads us to the following definition:

Let f be a function and x a value in the function's domain. We define the **derivative of f**, a new function called f', by the formula

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

$f'(x)$ is sometimes written as $\frac{d}{dx} f(x)$.

Example: The Derivative Function

Examples:

1. Find $f'(x)$ for $f(x) = x^2$. Use this function formula to find $f'(2)$.

Example: The Derivative Function

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1. Find $f'(x)$ for $f(x) = x^2$. Use this function formula to find $f'(2)$.
2. Find $g'(x)$ for $g(x) = 6$.

Example: The Derivative Function

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1. Find $f'(x)$ for $f(x) = x^2$. Use this function formula to find $f'(2)$.
2. Find $g'(x)$ for $g(x) = 6$.
3. Find $h'(x)$ for $h(x) = 4x - 2$.

Example: The Derivative Function

Examples:

1. Find $f'(x)$ for $f(x) = x^2$. Use this function formula to find $f'(2)$.
2. Find $g'(x)$ for $g(x) = 6$.
3. Find $h'(x)$ for $h(x) = 4x - 2$.
4. Find $\ell'(x)$ for $\ell(x) = ax^2 + bx + c$.

Example: The Derivative Function

Example: Let $f(x) = 3x^2 + 6$.

1. Use the derivative at a point formula to find $f'(-2)$.

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2. Use the derivative at a point formula to find $f'(6)$.
3. Use the definition of the derivative formula to find $f'(x)$.

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1. Use the derivative at a point formula to find $f'(-2)$.
2. Use the derivative at a point formula to find $f'(6)$.
3. Use the definition of the derivative formula to find $f'(x)$.
4. Find $f'(-2)$ with the derivative function.

Example: The Derivative Function

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2. Use the derivative at a point formula to find $f'(6)$.
3. Use the definition of the derivative formula to find $f'(x)$.
4. Find $f'(-2)$ with the derivative function.
5. Find $f'(6)$ with the derivative function.

Example: The Derivative Function

Example: Let $g(x) = \dfrac{1}{x}$.

1. Use the derivative at a point formula to find $g'(-3)$.

Example: The Derivative Function

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4. Find $g'(-3)$ with the derivative function.

Example: The Derivative Function

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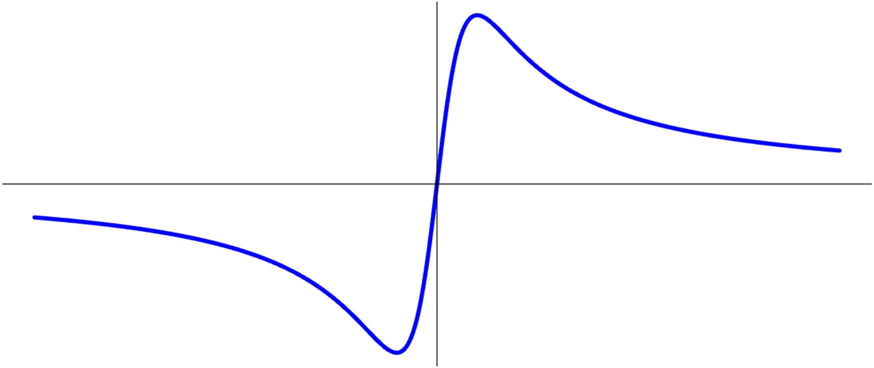
1. Use the derivative at a point formula to find $g'(-3)$.
2. Use the derivative at a point formula to find $g'(1)$.
3. Use the definition of the derivative formula to find $g'(x)$.
4. Find $g'(-3)$ with the derivative function.
5. Find $g'(1)$ with the derivative function.

The Derivative Function, Graphically

Example: Sketch the graph of $q'(x)$ if $q(x)$ has the graph below.



Application of the Derivative

Recall:

We say that f is **increasing on** (a, b) provided that the function is always rising as we move from left to right. That is, for any x and y in (a, b), if $x < y$, then $f(x) < f(y)$.

We say that f is **decreasing on** (a, b) provided that the function is always falling as we move from left to right. That is, for any x and y in (a, b), if $x < y$, then $f(x) > f(y)$.

Application of the Derivative

It can then be shown that:

Let f be a function that is differentiable on an interval (a, b). If $f'(x) > 0$ for every x such that $a < x < b$, then f is **increasing on (a, b)**.

Let f be a function that is differentiable on an interval (a, b). If $f'(x) < 0$ for every x such that $a < x < b$, then f is **decreasing on (a, b)**.

The Derivative Function, Graphically

Example: Given the graph of $r'(x)$ below, give a rough sketch of $r(x)$.